Paleographical and numerological results from a study of the original Unicode encoding of the Rosetta Stone

William Gunther
Google
wgunther@google.com

Brian Kell
Duolingo
brian.kell@duolingo.com

SIGBOVIK 2026
Carnegie Mellon University
10th April 1, 2026

Abstract

The Rosetta Stone is one of the most famous ancient Egyptian hieroglyphic texts in the world because of the secrets it has stubbornly refused to yield. In this paper we take the first steps toward the decipherment of these hidden secrets by applying rigorous numerological techniques to the original Unicode encoding of the hieroglyphic text. In doing so, we found that we actually had to *produce* the original Unicode encoding of the Rosetta Stone, along with a rendering engine that can display this Unicode encoding correctly, neither of which had ever been done before (surprisingly).

1 Introduction and background

The ancient Egyptian hieroglyphic script arose about five thousand years ago [50]. The way that this writing system worked, according to one eminent scholar, is that "people drew little pictures" [1]. Consequently, "the hieroglyphs were complex and writing them took a long time" [14]. But "[f]or a ling time, we didn't know what hieroglyphics said!" [1].

Ancient Egyptian hieroglyphs have been included in the Unicode standard since 2009, with significant extensions in 2022 and 2024. Hence each hieroglyph is associated with a numeric code point. Of course the ancient Egyptians must have been aware of the numerical values of their own writing system, so it is reasonable to suppose that at least some hieroglyphic texts have secret secondary messages encoded in these numbers. In this paper, we aim to discover these secrets using well established numerological techniques.

1.1 Character encoding for Egyptian hieroglyphics

The 1799 discovery of the Rosetta Stone [35] led to an interest in hieroglyphics in the first quarter of the 19th century by Champollion et al. [42, 36, 6, 38]. Since then, tens of thousands of hieroglyphic texts have been studied. However, until fairly recently this work has been done with an almost complete lack of consideration of the character encoding used by the ancient Egyptians.

This situation has begun to change over the past three decades. Significant discoveries have been made concerning the character encoding of ancient Egyptian hieroglyphs, beginning with the seminal papers of Everson in 1997 [9, 10]. Other important research includes additional work by Everson [11] as well as that of Bunz [3], Cook et al. [4], and Everson and Richmond [12, 13], culminating in the incorporation of Egyptian hieroglyphs in Unicode 5.2 [45] in 2009.

Since then study in this field has continued apace. In 2010, Minohara discovered that ancient Egyptian hieroglyphs are actually simplified Japanese [28]. In 2015, Davis [5] suggested a possible link between Egyptian hieroglyphs and emoji, but this connection appears not to have been borne out by further research.

Richmond [37] was the first to recognize that the arrangement of hieroglyphs into quadrats (clusters) was itself associated with code points. This intriguing new direction was explored further by, among others, Nederhof et al. [31, 32], Glass [16], and Glass et al. [17]. One particularly interesting result is that of

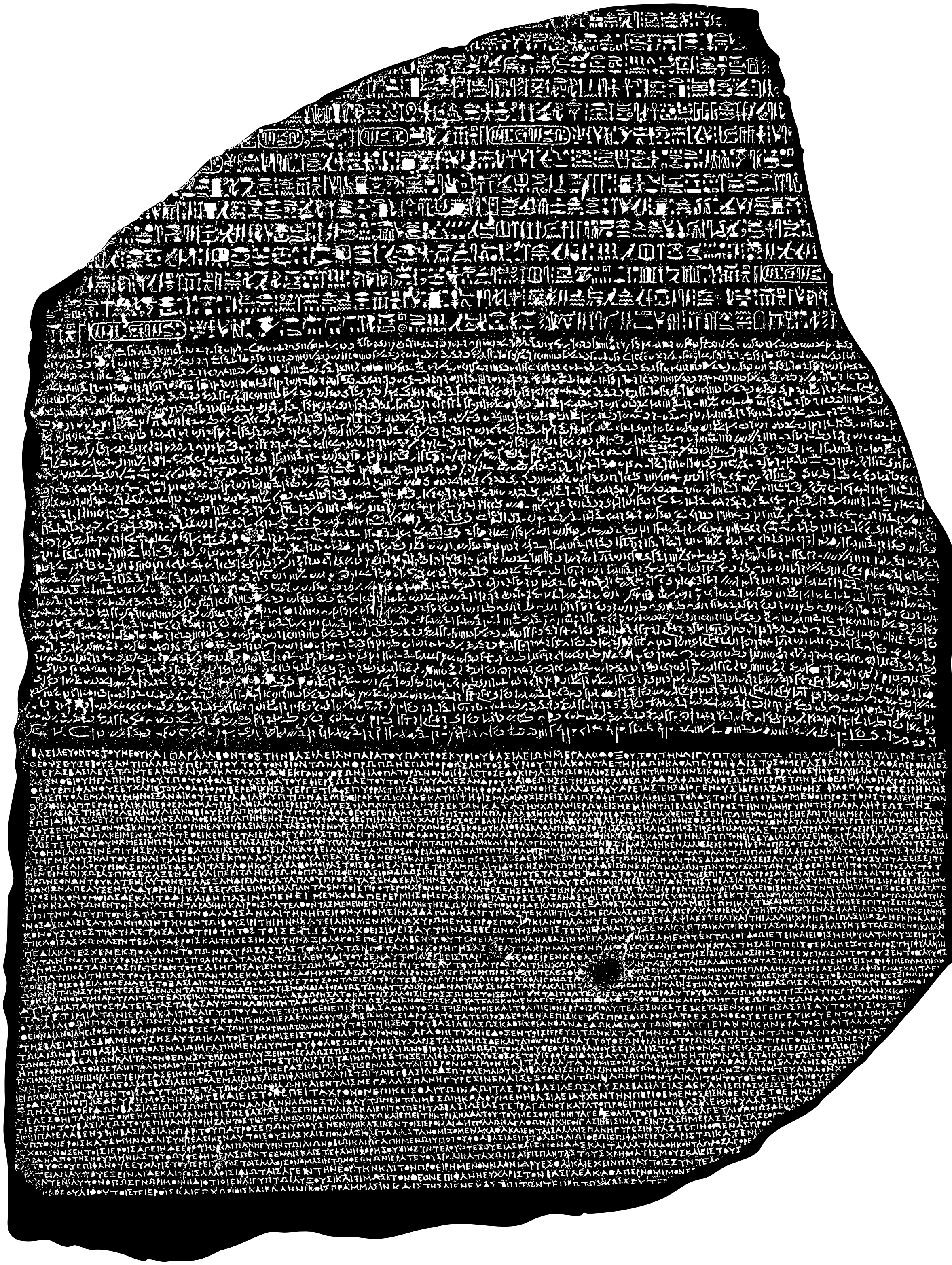

Figure 1: The Rosetta Stone [35, 24].

Nederhof [33], which demonstrated that ancient Egyptian parser generators must have supported $LR(k)$ grammars for $k \geq 2$. Hieroglyph format controls first appeared in Unicode 12.0 [46] in 2019. Glass et al. have continued work in this area [18, 19], which was adopted in Unicode 15.0 [47] in 2022.

Suignard has discovered the encodings of thousands of additional hieroglyphs, beginning with preliminary results in 2016 [43]; by 2023 these results had been generally accepted [44] and were included in Unicode 16.0 [48] in 2024.

1.2 Numerology

By the end of 2023, then, the numeric code point values of much of the Egyptian hieroglyphic script had been determined. Naturally one would expect these discoveries to have led to a deeper understanding of ancient Egyptian texts via standard numerological techniques. To the contrary, however, there has been surprisingly little work done in this direction.

In early 2024 the present authors decided to remedy this gap in order to have something to present at the illustrious SIGBOVIK '24 conference. The plan was threefold:

1. Obtain the Unicode version of a standard ancient Egyptian text, such as the aforementioned Rosetta Stone.
2. Analyze the code points of the text with industry-standard numerological techniques, such as isopsephy [27, 21], gematria [25], or Drosnin's method [8, 15].
3. Unlock the hidden secrets of the universe.

Unfortunately we got stuck almost immediately on step 1. We were surprised to find that there essentially are no Unicode versions of ancient Egyptian texts. It appears that most of the research that has been done since 1997 to work out the code points of Egyptian hieroglyphs has been entirely theoretical in nature; there has been a severe lack of practical application of these results.

In particular, at that time there was no software support for the Egyptian Hieroglyph Format Controls block [7, 40, 29]. As the proper arrangement of hieroglyphs into quadrats is important for the correct representation of ancient Egyptian texts, this lack of support seriously limited the usage of Unicode for hieroglyphics [37]. The challenge of separating the notions of *character* and *glyph* in ancient Egyptian texts is also a factor in the slow adoption of Unicode [30]. Some Egyptologists were urging the use of Unicode for hieroglyphic text as early as 2010 [20], but because of the lack of widespread software support even for basic Egyptian hieroglyphs, others were still recommending alternative approaches in 2022 [22]. The encoding of hieroglyphic texts in Unicode has even been declared an "impossibility" [23].

So, in 2024, we found ourselves stymied. The mysteries of ancient Egyptian hieroglyphics would have to wait for a future generation.

2 AI to the rescue

Fast-forward to 2026. With the advent of LLMs and widespread generative AI, we now have a promising new technology for attacking the problem. We don't need to have the Rosetta Stone already encoded in Unicode, nor do we need to have a pre-existing rendering engine that supports hieroglyph format controls. AI agents can enable us, two laymen with no knowledge or understanding of the ancient Egyptian language or writing system, to perfectly recreate the original Unicode representation of the Rosetta Stone text and build a rendering engine to display it properly.

3 Encoding and rendering pipeline

To properly analyze the hieroglyphic text, we first needed to determine the intended Unicode encoding of the Rosetta stone; Ancient Egyptians did not provide the text in a form from which this would be easy to extract (e.g., a direct sequence of Unicode code points). Therefore, this needs to be inferred from the rendering they created.

Fortunately, there has been some independent academic interest in the Rosetta Stone. The Rosetta Stone Online, a project in cooperation of the Excellence Cluster Topoi and the Institut für Archäologie, Humboldt-Universität zu Berlin, has a readily accessible XML encoding of the Rosetta Stone. Individual hieroglyphs are rendered using Unicode when available. The first challenge, however, is that there was no encoding that we could find that used the 2D spatial arrangement Unicode codepoints for Egyptian hieroglyphs. Even worse, there seemed to be no rendering tool that was capable of interpreting all these arrangement codepoints.

There was, however, an OpenType font (NewGardiner) and a rendering engine (`opentypehiero`) that did implement a lot of the glyphs and some of the 2D rendering (like vertical stacking). We forked this library and font, and made two main improvements.

We added support for right-to-left (RTL) rendering using the Unicode RTL override codes, and implemented proper handling and rendering for the BEGIN SEGMENT and END SEGMENT Unicode markers, as well as CARTOUCHE markers.

Furthermore, we hit some snags because some required signs are not yet in the standard hieroglyph blocks. We found that a preliminary 2016 proposal by Suignard [43]) was more complete and seemingly discovered the glyphs we needed, so those were added. The addition of our own Egyptian hieroglyphs like this may seem to be a drastic step, but we are simply following the advice of McDonald [26].

In the end were were able to successfully produce a Unicode encoding of the Rosetta Stone hieroglyphics and render this Unicode text properly (see Figure 2). To the best of our knowledge, this is the first time either of these things has ever been fully done.[1]

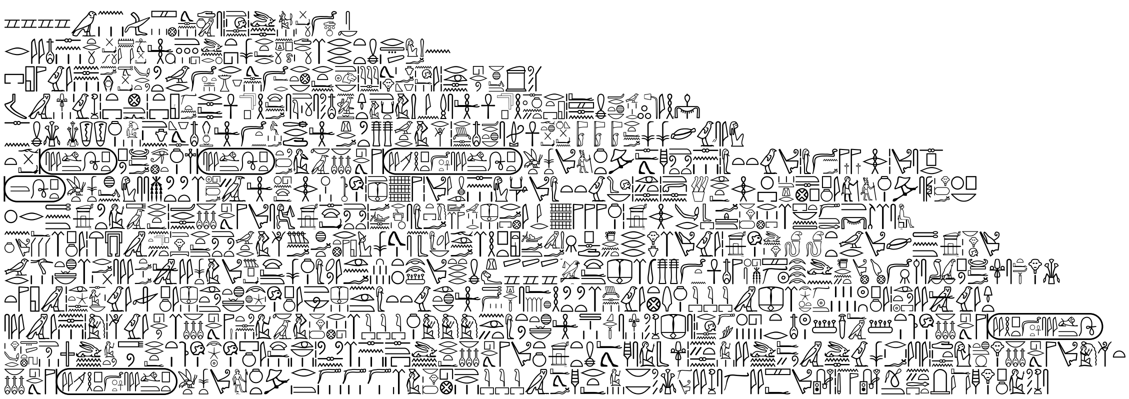

Figure 2: The hieroglyphic text of the Rosetta Stone, lines $x + 1$ to $x + 14$, rendered from Unicode using a fork of the NewGardiner font and `opentypehiero` library.

4 Observations

The ancient Egyptians appear to have restricted themselves to the use of hieroglyphs in the code point range 77824 to 82938. The significance of this constraint is not entirely clear, but we propose the following hypothesis. Obviously 82938 is the ZIP Code of McKinnon, Wyoming. Intriguingly, 77824 does not appear in the extant annals of the United States Postal Service [49], which suggests it was the ZIP Code for an ancient Egyptian city that has since been lost. The most likely candidate is Itj-Tawy, a capital city of the Twelfth Dynasty [41]. So the Egyptians probably chose this code point range for symbolic reasons, to connect their capital city with the famed glory of McKinnon, Wyoming. If this hypothesis is correct, we suggest, based on ZIP Code similarity, that archaeologists who are searching for the lost city of Itj-Tawy should focus their efforts in the vicinity of College Station, Texas.

[1] I know this is a SIGBOVIK paper and it's full of nonsense, but this is actually true.

5 Numerological investigations

Numerology may have been invented in Egypt, if it was not invented in China, Greece, Babylon, India, or Atlantis [2].

Previous work by Zauzich has focused on "hieroglyphs without mystery" [51], but here we are interested in hieroglyphs *with* mystery.

5.1 Isopsephy

Isopsephy, from the Greek *ísos* 'equal' and *psḗphos* 'pebble, counter, number,' is a Greek system that Menninger describes as the "numbering of letters" and the "lettering of numbers" [27]. To be more precise, Ifrah says that "isopsephy consists of determining the numerical value of a word or a group of letters, and relating it to another word by means of this value" [21].

Let us now take a bit of time to understand how isopsephy works. The 24 letters of the classical Greek alphabet, plus three obsolete letters named *digamma*, *koppa*, and *sampi*, each had an associated numerical value, as shown in Table 1. A word was then associated with a number by adding together the numerical values of its letters.

Units			Tens			Hundreds		
Α	alpha	1	Ι	iota	10	Ρ	rho	100
Β	beta	2	Κ	kappa	20	Σ	sigma	200
Γ	gamma	3	Λ	lambda	30	Τ	tau	300
Δ	delta	4	Μ	mu	40	Υ	upsilon	400
Ε	epsilon	5	Ν	nu	50	Φ	phi	500
Ϝ	digamma	6	Ξ	xi	60	Χ	chi	600
Ζ	zeta	7	Ο	omicron	70	Ψ	psi	700
Η	eta	8	Π	pi	80	Ω	omega	800
Θ	theta	9	Ϙ	koppa	90	Ϡ	sampi	900

Table 1: Numeric values of the Greek letters.

We shall give an example in just enough detail to convey the essential ideas. In Attic Greek, the name of Aristotle is written Ἀριστοτέλης; capitalized and without diacritics, this is ΑΡΙΣΤΟΤΕΛΗΣ. We observe that this name consists of eleven letters, which we may enumerate exhaustively as follows: the first letter is Α, the second letter is Ρ, the third letter is Ι, the fourth letter is Σ, the fifth letter is Τ, the sixth letter is Ο, the seventh letter is Τ, the eighth letter is Ε, the ninth letter is Λ, the tenth letter is Η, and the eleventh and final letter is Σ. The attentive reader will note that the fourth and eleventh letters are both Σ, and that the fifth and seventh letters are both Τ; we draw attention to these coincidences not because they are of any particular significance, but in the interest of completeness.

Now we can translate each of these letters to its corresponding numerical value according to Table 1: $\mathrm{A} \mapsto 1$, $\mathrm{P} \mapsto 100$, $\mathrm{I} \mapsto 10$, $\Sigma \mapsto 200$, $\mathrm{T} \mapsto 300$, $\mathrm{O} \mapsto 70$, $\mathrm{T} \mapsto 300$, $\mathrm{E} \mapsto 5$, $\Lambda \mapsto 30$, $\mathrm{H} \mapsto 8$, and $\Sigma \mapsto 200$. We pause to note that these eleven numerical values are not all distinct: the value 200 occurs twice, corresponding to the two instances of Σ noted above, and the value 300 occurs twice, corresponding to the two instances of Τ. The remaining seven values—namely 1, 100, 10, 70, 5, 30, and 8—each occur exactly once.

To derive the isopsephic value of the full name ΑΡΙΣΤΟΤΕΛΗΣ, then, we are required to add these eleven numerical values together. We do so in the order in which the corresponding letters appear in the name, though we note that addition being commutative, any other order would yield an identical result. We first add the values of the first two letters: $1 + 100 = 101$. Continuing, we incorporate the third letter: $101 + 10 = 111$. The fourth letter contributes a value of 200, giving $111 + 200 = 311$. The fifth letter contributes 300, giving $311 + 300 = 611$. The sixth letter contributes 70, giving $611 + 70 = 681$. The seventh letter, another Τ, contributes a further 300, giving $681 + 300 = 981$. The eighth letter contributes 5, giving $981 + 5 = 986$. The ninth letter contributes 30, giving $986 + 30 = 1016$. The tenth letter contributes 8, giving $1016 + 8 = 1024$. Finally, the eleventh letter, the second instance of Σ, contributes 200, giving $1024 + 200 = 1224$. We may verify this result by presenting the computation more compactly:

$$1 + 100 + 10 + 200 + 300 + 70 + 300 + 5 + 30 + 8 + 200 = 1224.$$

We thus conclude that the isopsephic value of ΑΡΙΣΤΟΤΕΛΗΣ is 1224.

Several additional examples may help to make the workings of the system clear. Ifrah [21] relates the following story, which has a serendipitous connection to ancient Egypt:

> In the *Pseudo-Callisthenes*† (I, 33) it is written that the Egyptian god Sarapis (whose worship was initiated by Ptolemy I) revealed his name to Alexander the Great in the following words:
>
> > Take two hundred and one, then a hundred and one, four times twenty, and ten. Then place the first of these numbers in the last place, and you will know which god I am.
>
> Taking the words of the god literally, we obtain
>
> 200 1 100 1 80 10 200
>
> which corresponds to the Greek name
>
> | Σ | Α | Ρ | Α | Π | Ι | Σ |
> |---|---|---|---|---|---|---|
> | 200 | 1 | 100 | 1 | 80 | 10 | 200 |
>
> - ›
>
> SARAPIS

According to Menninger [27], the most famous example of a number being associated with a (presumed) name is in the Book of Revelation, chapter 13, verse 18:

> Let him that hath understanding count the number of the beast: for it is the number of a man; and his number is Six hundred threescore and six—*kaì ho arithmòs autoû hexakósioi hexḗkonta héx*.

As another example, Ifrah [21] says that Suetonius (*Nero*, 39) equated the name *Nero* (in Greek, ΝΕΡΩΝ) to the phrase "he killed his own mother" (ΙΔΙΑΝ ΜΗΤΕΡΑ ΑΠΕΚΤΕΙΝΕ) because the letters in each add up to the same total in the Greek number system: ΝΕΡΩΝ is $50 + 5 + 100 + 800 + 50 = 1005$, while ΙΔΙΑΝ ΜΗΤΕΡΑ ΑΠΕΚΤΕΙΝΕ is $10+4+10+1+50+40+8+300+5+100+1+1+80+5+20+300+5+10+50+5 = 1005$.

This last example demonstrates an important feature of isopsephy, namely, the identification of a meaningful connection between two words or phrases if their numerical values are the same.

Unfortunately, we found it impossible to apply isopsephy to the Rosetta Stone, as it applies only to Greek and we were unable to find a Greek translation of the Rosetta Stone.

5.2 Gematria

Gematria is a numerological technique that takes a totally different approach from isopsephy. For starters, it uses the Hebrew alphabet, not the Greek one. This is a good characteristic for a numerological method to be applied to the Rosetta Stone, since the Rosetta Stone is written right to left and hieroglyphics omit most vowels.

Then, as Laitman [25] explains: "In Hebrew, each letter of the 22-letter alphabet is assigned a specific number. The first nine letters are assigned the numbers 1–9, respectively, and the next nine are assigned the numbers 10–90, with each letter jumping by 10 in value. The last four letters of the Hebrew alphabet are assigned the numbers 100, 200, 300, and 400, respectively." See Table 2.

† A spurious work associated with the name of Callisthenes, companion of Alexander in his Asiatic expedition.

Units			Tens			Hundreds		
א	alef	1	י	yod	10	ק	qof	100
ב	bet	2	כ	kaf	20	ר	resh	200
ג	gimel	3	ל	lamed	30	ש	shin	300
ד	dalet	4	מ	mem	40	ת	tav	400
ה	he	5	נ	nun	50			
ו	vav	6	ס	samekh	60			
ז	zayin	7	ע	ayin	70			
ח	het	8	פ	pe	80			
ט	tet	9	צ	tsadi	90			

Table 2: Numeric values of the Hebrew letters.

Another key difference from isopsephy is that in gematria the numeric value of a word is determined by summing up the numeric values of its letters, in contrast to isopsephy, in which the numerical values of the letters are added together to find the numerical value of the word.

Laitman [25] highlights yet another distinction from isopsephy: “There is significance when words include or add up to the same numbers; the meanings of the words that share numbers are thought to be related in the deepest sense.”

These properties make gematria, unlike isopsephy, uniquely suited for a text such as the Rosetta Stone. But it may be *too* good a match. We do not believe the Egyptians would have encoded their hidden messages using such an obvious method. In any case the SIGBOVIK paper deadline was rapidly approaching and the AI we used to vibe-code a numerology script did not think of Hebrew gematria for some reason, so we did not spend time pursuing this approach.

5.3 Drosnin’s method

Drosnin [8] popularized a method based on equidistant letter sequences, described succinctly by Gardner [15]. An equidistant letter sequence, or ELS, is a subsequence of a text formed by reading every nth character for some $n \geq 1$ (called the *stride*). The starting character may be chosen to be any of the first $n-1$ characters; this is called the *offset*, with the first character of the whole text being offest 0. Then words and phrases are sought within each ELS.

5.3.1 Methodology

The first step of applying this method to the Egyptian hieroglyphs on the Rosetta stone is to linearize them into a stream of Unicode code points, which we now have.

One challenge that presents itself is the problem of interpreting a given ELS. Applying Drosnin’s method directly to the hieroglyphic text produces a subsequence of hieroglyphs, which of course is garbage—no one can read that. Instead we must translate the hieroglyphs into something meaningful, such as the letters of the English language.

There are several possible conversion methods that can be used to map hieroglyphs to English letters. Mathematical approaches are an obvious possibility. For example, one can simply take the Unicode code point value mod 26. Or the bytes of the code point value can be manipulated first, such as by XORing the two bytes before the mod-26 operation. The conversion method may make use of state, such as by taking the absolute value of the difference between the current code point and the previous code point, and then performing a mod-26 operation on the result.

Alternatively, we may make use of an exact equivalence between English letters and certain hieroglyphs, as discovered by Arlon [1], shown in Table 3. Clearly if we see the hieroglyph 𓄿, which has code point value 78143, this should be interpreted as the letter A. If we see a hieroglyph with a code point value close to this, such as 78144 or 78141, we can assume that this is just a typo, and the scribe or stone carver probably meant to make an A. This suggests an approach based on one-dimensional Voronoi regions centered at the hieroglyphs in Table 3.

Letter	**Hieroglyph**	**Letter**	**Hieroglyph**	**Letter**	**Hieroglyph**	**Letter**	**Hieroglyph**
A	𓄿	H	𓉔	O	𓍯	V	𓆑
B	𓃀	I	𓇋	P	𓊪	W	𓅱
C	𓎡	J	𓆓	Q	𓈎	X	𓐍
D	𓂧	K	𓎡	R	𓂋	Y	𓇌
E	𓂝	L	𓃭	S	𓋴	Z	𓊃
F	𓆑	M	𓅓	T	𓏏		
G	𓎼	N	𓈖	U	𓅱		

Table 3: Exact hieroglyphic equivalents of English letters.

(An observant reader may notice that some pairs of letters are associated with the same hieroglyph: C and K, F and V, and U and W. We construe this to mean that the Ancient Egyptians were poor spellers, and sometimes a given hieroglyph may have multiple interpretations.)

Of course, the Egyptians were unlikely to hide secret messages in their monumental inscriptions in plaintext. They probably used some form of encryption, likely a Caesar cipher, considering the close relationship between Caesar and Cleopatra [39].

5.3.2 Results

With these techniques we found several intriguing messages hidden in the Unicode text of the Rosetta Stone.

First, an important early success that confirmed we were on the right track was the discovery that the word TRUTH appears three times under the simplest mod-26 conversion, even with no Caesar shift: at stride 14, offset 4, position 97; stride 70, offset 48, position 18; and stride 219, offset 149, position 2.

Various conversion methods, Caesar shifts, strides, and offsets yield words such as CLOAKAGE, DUCTLESS, TABULATA, DESISTED, COCOWOOD, BOWLLIKE, THADDEUS, STOVEMAN, and SELFLESS. The Voronoi conversion method based on Table 3 produces EXACT, FATWA, KHASA, OVENLY, MALIK, TAXWAX, and TITOIST.

But perhaps the most striking result comes from examining the equidistant letter sequences with stride 60, using the difference-from-previous-code-point conversion method. At offset 38 we can find the word DRINK; at offset 31 we can find the word YOUR. The top part of the Rosetta stone has been broken off and lost, obscuring the rest of the message, but the surviving fragment is enough to reconstruct the full message: BE SURE TO DRINK YOUR OVALTINE.

6 Conclusions and future work

We clearly started this work too late to get significant numerological results before the SIGBOVIK paper deadline. But we can offer the world, for the first time, a correct and authoritative Unicode representation of the Rosetta Stone hieroglyphic text, as well as a Unicode rendering engine that displays it properly. We encourage other Egyptologists and numerologists to carry this work forward.

Recent work by Pallán [34] has focused on the Unicode code points of Mayan hieroglyphs. Preliminary results indicate that the Maya preferred hieroglyphs in the code point range 86016 to 87112, about 5–10% larger than the range favored by the Egyptians. The significance of this difference remains to be explored, but it suggests that there is wider variability than previously suspected in code points used by pyramid-building languages.

References

[1] Arlon, Penelope. 2014. *Ancient Egypt: Face-to-Face with Pharaohs, Gods, and Egypt's Everyday People: Bonus Amazing Mummy Tales*. Scholastic, Inc., New York. ISBN 978-0-545-62739-9.

[2] Atwood, Megan. 2019. *Numerology*. Compass Point Books, North Mankato. ISBN 978-0-7565-6103-1.

[3] Bunz, Carl-Martin. 2000. "Scripts from the past in future versions of Unicode®." 16th International Unicode Conference, Amsterdam. `https://www.unicode.org/L2/L2000/00128-pastscripts.pdf`.

[4] Cook, Richard; Everson, Michael; McGowan, Rick; and Richmond, Robert. 2005. "Revised proposal to encode Egyptian hieroglyphs in Plane 1 of the UCS." Unicode Technical Committee, document L2/05-311. `https://www.unicode.org/L2/L2005/05311-EgyptianHieroglyph.pdf`.

[5] Davis, M. 2015. "Hieroglyphs as emoji?" Unicode Technical Committee, document L2/15-208. `https://www.unicode.org/L2/L2015/15208-hieroglyphs-as-emoji.pdf`.

[6] Downs, Jonathan. 2008. *Discovery at Rosetta: The Stone That Unlocked the Mysteries of Ancient Egypt*. Skyhorse Publishing, New York. ISBN 978-1-602-39271-7.

[7] Draconis. 2021. Answer to "How do Egyptologists layout hieroglyphs in print?" *Linguistics*, Stack Exchange, New York. `https://linguistics.stackexchange.com/a/38680`.

[8] Drosnin, Michael. 1997. *The Bible Code*. Simon & Schuster, New York. ISBN 0-684-81079-4.

[9] Everson, Michael. 1997. "Encoding Egyptian hieroglyphs in ISO/IEC 10646-2." Unicode Technical Committee, document L2/97-266. `https://www.unicode.org/L2/L1997/97266-n1636.pdf`.

[10] Everson, Michael. 1997. "Proposal to encode Basic Egyptian Hieroglyphs in Plane 1." Unicode Technical Committee, document L2/97-267. `https://www.unicode.org/L2/L1997/97267-n1637.pdf`.

[11] Everson, Michael. 1999. "Encoding Egyptian hieroglyphs in Plane 1 of the UCS." ISO/IEC JTC1/SC2/WG2 N1944. `https://www.unicode.org/wg2/docs/n1944.pdf`.

[12] Everson, Michael, and Richmond, Bob. 2006. "Towards a proposal to encode Egyptian hieroglyphs in the UCS." ISO/IEC JTC1/SC2/WG2 N3181. `https://www.unicode.org/wg2/docs/n3181.pdf`.

[13] Everson, Michael, and Richmond, Bob. 2007. "Proposal to encode Egyptian hieroglyphs in the SMP of the UCS." ISO/IEC JTC1/SC2/WG2 N3237. `https://www.unicode.org/wg2/docs/n3237.pdf`.

[14] Fullman, Joseph. 2013. *Eyewitness: Ancient Civilizations*. DK Publishing, New York. ISBN 978-1-4654-0888-4.

[15] Gardner, Martin. 2000. *Did Adam and Eve Have Navels?: Discourses on Reflexology, Numerology, Urine Therapy, and Other Dubious Subjects*. W. W. Norton & Company, New York. ISBN 0-393-04963-9.

[16] Glass, Andrew. 2016. "Preliminary analysis of Egyptian Hieroglyph quadrat types." Unicode Technical Committee, document L2/16-232. `https://www.unicode.org/L2/L2016/16232-quadrat-types.pdf`.

[17] Glass, Andrew; Hafemann, Ingelore; Nederhof, Mark-Jan; Polis, Stéphane; Richmond, Bob; Rosmorduc, Serge; and Schweitzer, Simon. 2017. "A method for encoding Egyptian quadrats in Unicode." ISO/IEC JTC1/SC2/WG2 N4818. `https://www.unicode.org/wg2/docs/n4818-quadrat-encoding.pdf`.

[18] Glass, Andrew; Grotenhuis, Jorke; Nederhof, Mark-Jan; Polis, Stéphane; Rosmorduc, Serge; and Werning, Daniel A. 2021. "Additional control characters for Ancient Egyptian hieroglyphic texts." Unicode Technical Committee, document L2/21-208. `https://www.unicode.org/L2/L2021/21208-egyptian-ctrl.pdf`.

[19] Glass, Andrew; Grotenhuis, Jorke; Nederhof, Mark-Jan; Polis, Stéphane; Rosmorduc, Serge; and Werning, Daniel A. 2021. "Additional control characters for Ancient Egyptian hieroglyphic texts." Unicode Technical Committee, document L2/21-248. `https://www.unicode.org/L2/L2021/21248-egyptian-controls.pdf`.

[20] Gozzoli, Roberto. 2010. "Hieroglyphic text processors, Manuel de Codage, Unicode, and lexicography." *Texts, Languages & Information Technology in Egyptology: Selected papers from the meeting of the Computer Working Group of the International Association of Egyptologists (Informatique & Égyptologie), Liège, 6–8 July 2010*, Stéphane Polis and Jean Winand, eds., Presses Universitaires de Liège, 2013, pp. 89–101.

[21] Ifrah, Georges. 2000. *The Universal History of Numbers: From Prehistory to the Invention of the Computer*. John Wiley & Sons, Inc., New York. ISBN 0-471-37568-3. Translated from the French by David Bellos, E. F. Harding, Sophie Wood, and Ian Monk.

[22] Jauhiainen, Heidi. 2022. "Encoding hieroglyphic texts." 6th Digital Humanities in the Nordic and Baltic Countries Conference (DHNB 2022), Uppsala, pp. 244–250.

[23] Jauhiainen, Heidi. 2024. "Gly2Mdc v.2.0: Lessons learned from building a tool for hieroglyphic texts." 2024 Digital Humanities in the Nordic and Baltic Countries (DHNB), Reykjavík.

[24] Kinetoons. 2014. "The Rosetta Stone." Openclipart. `https://openclipart.org/204962`, `https://commons.wikimedia.org/wiki/File:Rosetta_Stone.svg`.

[25] Laitman, Rav Michael, Ph.D., with Canright, Collin. 2007. *The Complete Idiot's Guide to Kabbalah*. Penguin Group, New York. ISBN 978-1-59257-542-8.

[26] McDonald, Angela. 2007. *Write Your Own Egyptian Hieroglyphs*. University of California Press, Berkeley. ISBN 978-0-520-25235-6.

[27] Menninger, Karl. 1969. *Number Words and Number Symbols: A Cultural History of Numbers*. M.I.T. Press, Cambridge. ISBN 0-262-13040-0. Translated from the revised German edition by Paul Broneer.

[28] Minohara, Tatsuo. 2010. "A writing system for the ancient Egyptian hieroglyphs." 7th International Conference on Informatics and Systems (INFOS), Cairo, pp. 1–7.

[29] Napier, Rob. 2023. Answer to "Unicode Egyptian cartouche." Stack Overflow, New York. `https://stackoverflow.com/a/76590852`.

[30] Nederhof, Mark-Jan. 2010. "The Manuel de Codage encoding of hieroglyphs impedes development of corpora." *Texts, Languages & Information Technology in Egyptology: Selected papers from the meeting of the Computer Working Group of the International Association of Egyptologists (Informatique & Égyptologie), Liège, 6–8 July 2010*, Stéphane Polis and Jean Winand, eds., Presses Universitaires de Liège, 2013, pp. 103–110.

[31] Nederhof, Mark-Jan; Rajan, Vinodh; Lang, Johannes; Polis, Stéphane; Rosmorduc, Serge; Richter, Tonio Sebastian; Hafemann, Ingelore; and Schweitzer, Simon. 2016. "A comprehensive system of control characters for Ancient Egyptian hieroglyphic text (preliminary version)." Unicode Technical Committee, document L2/16-177. `https://www.unicode.org/L2/L2016/16177-egyptian.pdf`.

[32] Nederhof, Mark-Jan; Rajan, Vinodh; Lang, Johannes; Polis, Stéphane; Rosmorduc, Serge; Richter, Tonio Sebastian; Hafemann, Ingelore; and Schweitzer, Simon. 2017. "A system of control characters for Ancient Egyptian hieroglyphic text (updated version)." Unicode Technical Committee, document L2/16-210R. `https://www.unicode.org/L2/L2016/16210r-egyptian-control.pdf`.

[33] Nederhof, Mark-Jan. 2018. "A note on the syntax of Ancient Egyptian hieroglyphic control characters." Unicode Technical Committee, document L2/18-236. `https://www.unicode.org/L2/L2018/18236-nederhof.pdf`.

[34] Pallán Gayol, Carlos. 2023. "Updated list of characters for Mayan codices." Unicode Technical Committee, document L2/23-228. `https://www.unicode.org/L2/L2023/23228-mayan-update_korr.pdf`.

[35] Priests of Pharaoh Ptolemy V Epiphanes. 196 B.C. *Rosetta Stone*. Granodiorite Stele, Memphis.

[36] Ray, John. 2007. *The Rosetta Stone and the Rebirth of Ancient Egypt*. Harvard University Press, Cambridge. ISBN 978-0-674-02493-9.

[37] Richmond, Bob. 2015. "Proposal to encode three control characters for Egyptian Hieroglyphs." Unicode Technical Committee, document L2/15-123. `https://www.unicode.org/L2/L2015/15123r-egyptian.pdf`.

[38] Robinson, Andrew. 2012. *Cracking the Egyptian Code: The Revolutionary Life of Jean-François Champollion*. Oxford University Press, Oxford. ISBN 978-0-19-991499-9.

[39] Shaw, George Bernard. 1898. "Caesar and Cleopatra." In *Three Plays for Puritans*. Grant Richards, London.

[40] Skomisa. 2022. Answer to "Using the Egyptian Hieroglyphic control characters." Stack Overflow, New York. `https://stackoverflow.com/a/72579104`.

[41] Snape, Steven. 2014. *The Complete Cities of Ancient Egypt*. Thames & Hudson Ltd, London. ISBN 978-0-500-05179-5.

[42] Solé, Robert, and Valbelle, Dominique. 2002. *The Rosetta Stone: The Story of the Decoding of Hieroglyphics*. Four Walls Eight Windows, New York. ISBN 1-56858-226-9. Translated by Steven Rendall.

[43] Suignard, Michel. 2016. "Preliminary draft for the encoding of an extended Egyptian Hieroglyphs repertoire." Unicode Technical Committee, document L2/16-079. `https://www.unicode.org/L2/L2016/16079-hieroglyphs.pdf`.

[44] Suignard, Michel; Anderson, Deborah; Dils, Peter; Grotenhuis, Jorke; Gülden, Svenja A.; Nederhof, Mark-Jan; Polis, Stéphane; Richmond, Robert; Rosmorduc, Serge; and Werning, Daniel A. 2023. "Encoding proposal for an extended Egyptian Hieroglyphs repertoire." ISO/IEC JTC1/SC2/WG2 N5240R2. `https://www.unicode.org/wg2/docs/n5240R2-hieroglyphs.pdf`.

[45] Unicode Consortium, The. 2009. *The Unicode Standard, Version 5.2*. The Unicode Consortium, Mountain View, ISBN 978-1-936213-00-9. `http://www.unicode.org/versions/Unicode5.2.0/`.

[46] Unicode Consortium, The. 2019. *The Unicode Standard, Version 12.0.0*. The Unicode Consortium, Mountain View, ISBN 1-936213-22-1. `http://www.unicode.org/versions/Unicode12.0.0/`.

[47] Unicode Consortium, The. 2022. *The Unicode Standard, Version 15.0.0*. The Unicode Consortium, Mountain View, ISBN 978-1-936213-32-0. `https://www.unicode.org/versions/Unicode15.0.0/`.

[48] Unicode Consortium, The. 2024. *The Unicode Standard, Version 16.0.0*. The Unicode Consortium, South San Francisco, ISBN 978-1-936213-34-4. `https://www.unicode.org/versions/Unicode16.0.0/`.

[49] United States Postal Service. 2026. *Look Up a ZIP Code™*. `https://tools.usps.com/zip-code-lookup.htm?citybyzipcode`.

[50] Wilson, Penelope. 2003. *Hieroglyphs: A Very Short Introduction*. Oxford University Press, Oxford. ISBN 978-0-19-280502-7.

[51] Zauzich, Karl-Theodor. 1992. *Hieroglyphs Without Mystery*. University of Texas Press, Austin. ISBN 0-292-79804-0. Translated and adapted by Ann Macy Roth from *Hieroglyphen ohne Geheimnis*, 1980, Verlag Philipp von Zabern, Mainz.